

JOY DATTA

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RESEARCH INTERESTS

AI Coding, Self-supervised Representation Learning, Generative AI, Machine Learning Applications

EDUCATION

University of Iowa August 2026 - Present

PhD in Computer Science (Advisor: [Prof. Peng Jiang](#))

BRAC University May 2017 - Jan 2022

BSc in Computer Science CGPA: **3.36/4.00**

Thesis: *Deep Convolutional GAN-based Data Augmentation for Medical Image Classification* [\[Open Access\]](#)

Advisor: [Moin Mostakim](#) **Thesis Grade:** 4.0/4.0

Relevant Coursework: Artificial Intelligence, Neural Networks, Linear Algebra, Differential & Integral Calculus, Statistics & Probability, Stochastic Processes, Discrete Mathematics, Introduction to Robotics.

RESEARCH EXPERIENCE

Graduate Research Assistant (*Supervised by* [Prof. Peng Jiang](#)) Aug 2026 - Present

Department of Computer Science, University of Iowa

Research Assistant (*Supervised by* [Prof. Jia Uddin](#)) Jan 2023 - Apr 2026

Multimedia Signal and Image Processing Research Group

Research Areas: Representation Learning, Resource-Efficient Models, Spiking Neural Networks

- Explored various strategies to visual representation learning with limited use of backpropagation and biologically plausible learning, including layer-wise training of deep models via self-supervised techniques.
- Investigated the parameter-efficiency of convolutional spiking neural networks.

PUBLICATIONS

JOURNAL ARTICLES

[1] Deep representation learning using layer-wise VICReg losses *Scientific Reports* (SCIE indexed (**Q1**), Journal Impact Factor - 3.9)

J. Datta, R. Rabbi, P. Saha, A. N. Zereen, M. Abdullah-Al-Wadud, and J. Uddin, “Deep representation learning using layer-wise VICReg losses,” *Scientific Reports*, vol. 15, no. 1, Art. no. 27049, 2025. [\[Open Access\]](#)

Funded by: King Saud University, Riyadh, Saudi Arabia, Ongoing Research Funding program (ORF-2025-951).

CONFERENCE PAPAERS

[1] Minimizing the Effect of Sleep Deprivation in the Forward-Forward Algorithm *ICPR 2026*

J. Datta, P. Saha, R. Rabbi, N.I. Rafin, S. Shatabda, M.G.R. Alam, and C. Mourning, “Minimizing the Effect of Sleep Deprivation in the Forward-Forward Algorithm,” in *28th International Conference on Pattern Recognition (ICPR)*, Lyon, France, Aug. 17-22, 2026, pp. 590-604. [\[Paper\]](#)

[2] POM-NeRF: Perception-Oriented Modification of Neural Radiance Fields for Enhanced 3D Volumetric Rendering from Multi-View Images *VISAPP 2026*

J. Datta, R. Rabbi, M. S. I. Tonmoy, P. Saha, and C. Mourning, “POM-NeRF: Perception-Oriented Modification of Neural Radiance Fields for Enhanced 3D Volumetric Rendering from Multi-View Images,” in *21st. International Conference on Computer Vision Theory and Applications (VISAPP)*, Marbella, Spain, Mar. 9-11, 2026, to be published. [[Open Access](#)]

[3] Meta-Ensemble of FastFlow and PaDiM for Efficient Industrial Anomaly Detection *RAAICON 2025*

P. Saha, **J. Datta**, S. Dey, and J. Rahman, “Meta-Ensemble of FastFlow and PaDiM for Efficient Industrial Anomaly Detection,” in *2025 IEEE 4th International Conference on Robotics, Automation, Artificial-Intelligence and Internet-of-Things (RAAICON)*, Dhaka, Bangladesh, Nov. 27–28, 2025. [[IEEE Xplore](#)]

[4] Peer-Guided Optimization: Incorporating Collaborative Learning into Stochastic Optimization in Machine Learning *ECCE 2025*

J. Datta, R. Rabbi, N. I. Rafin, and M. G. R. Alam, “Peer-Guided Optimization: Incorporating Collaborative Learning into Stochastic Optimization in Machine Learning,” in *2025 International Conference on Electrical, Computer and Communication Engineering (ECCE)*, Chittagong, Bangladesh, Feb. 13–15, 2025, pp. 1–6. [[IEEE Xplore](#)]

[5] Parameter-Efficient Image Classification with Convolutional Spiking Neural Network via Fast Sigmoid Surrogate Gradient Descent *ECCE 2025*

J. Datta, F. A. Sarwar, R. Rabbi, P. Saha, M. G. R. Alam, J. Uddin, and A. N. Zereen, “Parameter-Efficient Image Classification with Convolutional Spiking Neural Network via Fast Sigmoid Surrogate Gradient Descent,” in *2025 International Conference on Electrical, Computer and Communication Engineering (ECCE)*, CUET, Chittagong, Bangladesh, Feb. 13–15, 2025, pp. 1–6. [[IEEE Xplore](#)]

[6] Wildfire Prediction using Convolutional Kolmogorov-Arnold Network *ICECE 2024*

R. Rabbi*, **J. Datta***, S. Ahmed, and A. N. Zereen, “Wildfire Prediction using Convolutional Kolmogorov-Arnold Network,” in *13th International Conference on Electrical and Computer Engineering (ICECE)*, Dhaka, Bangladesh, Dec. 18–20, , pp. 667-672. [[IEEE Xplore](#)]

*Contributed equally.

SELECTED PROJECTS

- Visualizing how the representation spaces evolve in a deep model during training with tSNE [[code](#)]
- Restoring blur images using denoising convolutional autoencoder [[code](#)]
- Image generation with Variational Autoencoder and its latent space visualization [[code](#)]

TECHNICAL SKILLS

Programming Languages: Python, Java, SQL

Machine Learning Tools: Tensorflow, Keras, Pytorch, Scikit-learn, OpenCV

Research & Developer Tools: JupyterLab, Jupyter Notebook, VSCode, Spyder, Sublime Text, Git

Data Analysis & Visualization: Numpy, Pandas, Matplotlib, Seaborn

Miscellaneous: OOP, Data Structures, Vector Databases, RAG, LangChain, LangGraph

Writing & Presentation: Google Docs & Slides, Microsoft Word & Powerpoint, L^AT_EX

ACHIEVEMENTS

VC's List, BRACU

Spring 2020

Dean's List, BRACU

Summer 2018

EXTRACURRICULAR ACTIVITIES

Robotics Club of BRAC University (ROBU)

Former Member (Creative Department)